

RAK14000 E-Ink Display Module Datasheet

Overview

Description

The **RAK14000 WisBlock E-Ink Display Module** is designed to be part of a production-ready IoT solution in a modular way and must be combined with a WisBlock Core and a WisBlock Base module.

The **RAK14000** module provides an interface to connect an E-Ink Display, which is known to be an extremely power-efficient display. The **E-Ink Display** is an Active Matrix Electrophoretic Display (AM EPD) module with 2.13 inches active area (212 × 104 pixels for White Black Version and 250 × 122 pixels for the White-Black-Red Version). It also allows you to connect a Three-Button Module, which can be used on a different application. All of this can be controlled through a WisBlock Core.

NOTE

There are two versions of RAK14000 module for sale in the RAKwireless Store, the [white-black display panel](#)  and [white-black-red display panel](#)  version.

Features

- Input Voltage: 3.3 V
- Display Size: 2.13 inches
- Wide Viewing Angles
- Pure Reflective Mode. Very viewable displays!
- On-chip display RAM and oscillator

- Extremely Low Power Consumption! Perfect for power-conscious applications.
- Comes with a 3-Button Module. Application can be customized.
- Uses FPC (Flexible PCB) for its Display and Button modules
- Chipset: Solomon Systech Limited SSD1680
- **White-Black**
 - Display Resolution: 212 × 104 pixels
- **White-Black-Red**
 - Display Resolution: 250 × 122 pixels

Specifications

Overview

Mounting

The **RAK14000 E-Ink Display Module** is mounted to the IO Slot of a WisBlock Base Board. Figure 1 shows the mounting mechanism of the RAK14000 on a WisBlock Base, such as the RAK5005-O.

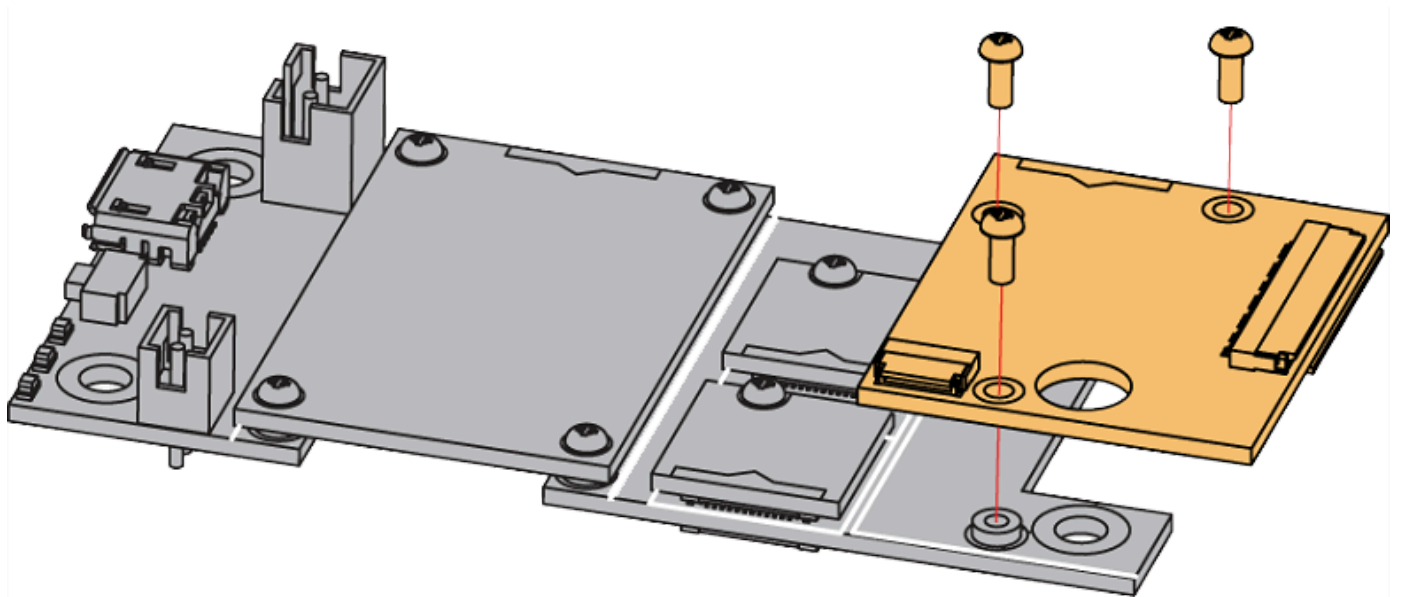


Figure 1: RAK14000 Module Mounting

Hardware

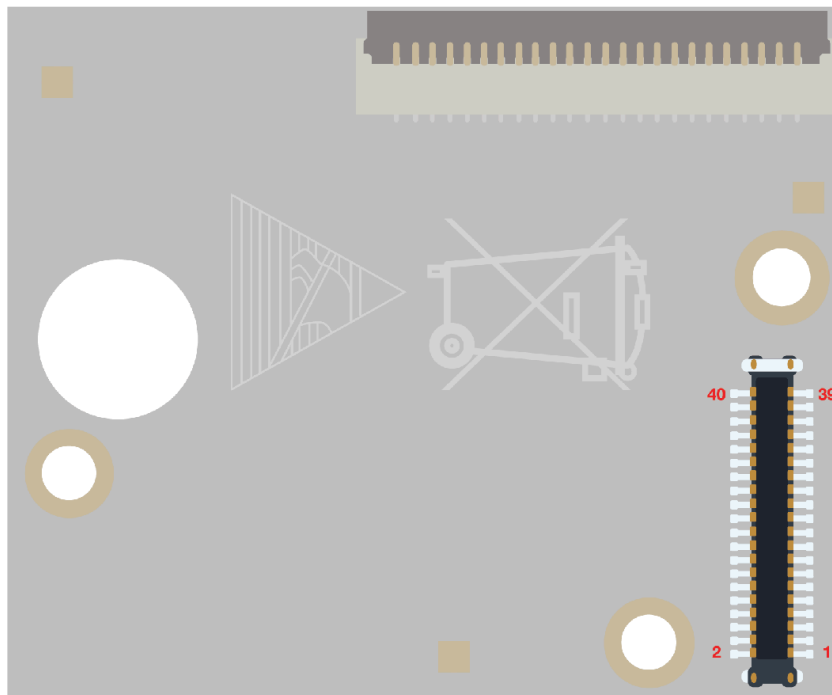
The hardware specification covers the pinouts of the board and its functionalities and diagrams. It also presents the parameters and their standard values.

Chipset

Vendor	Part number
Solomon Systech Limited	SSD1680

Pin Definition

The **RAK14000 WisBlock E-Ink Display Module** comprises a standard 40-pin WisConnector. The 40-pin WisConnector allows the RAK14000 module to be mounted on a WisBlock Base board, such as RAK5005-O. The pin order of the connector and the pinout definition is shown in Figure 2.



NC	2	1	NC
GND	4	3	GND
3v3	6	5	NC
NC	8	7	NC
NC	10	9	NC
NC	12	11	NC
NC	14	13	EINK_RESET
NC	16	15	NC
NC	18	17	NC
NC	20	19	NC
NC	22	21	NC
NC	24	23	NC
EINK_CSK	26	25	EINK_CS
EINK_SDIN	28	27	NC
NC	30	29	NC
EINK_BUSY	32	31	S1
NC	34	33	NC
NC	36	35	NC
S3	38	37	S2
GND	40	39	GND

Figure 2: RAK14000 Module Pinout

Pin Number	Pin Name	Description	Pin Number	Pin Name	Description
2	NC	Not Connected	1	NC	Not Connected
4	GND	Ground	3	GND	Ground
6	3v3	3.3V	5	NC	Not Connected
8	NC	Not Connected	7	NC	Not Connected
10	NC	Not Connected	9	NC	Not Connected

Pin Number	Pin Name	Description	Pin Number	Pin Name	Description
12	NC	Not Connected	11	NC	Not Connected
14	NC	Not Connected	13	EINK_RESET	EPD Reset signal. Active Low.
16	NC	Not Connected	15	NC	Not Connected
18	NC	Not Connected	17	NC	Not Connected
20	NC	Not Connected	19	NC	Not Connected
22	NC	Not Connected	21	NC	Not Connected
24	NC	Not Connected	23	NC	Not Connected
26	EINK_CSK	Serial Clock	25	EINK_CS	Chip select. Active Low.
28	EINK_SDIN	Serial Data	27	EINK_SDIN	Serial Data
30	NC	Not Connected	29	D/C#	Data /Command control
32	EINK_BUSY	Busy state	31	S1	Button Signal 1

Pin Number	Pin Name	Description	Pin Number	Pin Name	Description
34	NC	Not Connected	33	NC	Not Connected
36	NC	Not Connected	35	NC	Not Connected
38	S3	Button Signal 3	37	S2	Button Signal 2
40	GND	Ground	39	GND	Ground

NOTE

SPI_CS, SPI_MISO, SPI_MOSI, SPI_CLK, IO1, IO3-IO7 are connected to 40-pin WisConnector.

Electrical Characteristics

DC Characteristics

The following specifications apply for: VSS = 0V, VCI = 3.0 V, TOPR = 25 °C

Symbol	Description	Min.	Nom.	Max.	Unit
V _{CI}	Logic supply voltage	2.2	3.0	3.7	V
V _{DD}	Core logic voltage	1.7	1.8	1.9	V
I _{slp}	Sleep mode current	-	20	-	μA

Symbol	Description	Min.	Nom.	Max.	Unit
I_{dslp}	Deep sleep mode current	-	1	5	μA
I_{opr}	Typical operating current	-	4.3	4.5	mA

Mechanical Characteristics

Figure 3 shows the dimensions and mechanical drawing of the RAK14000 module.

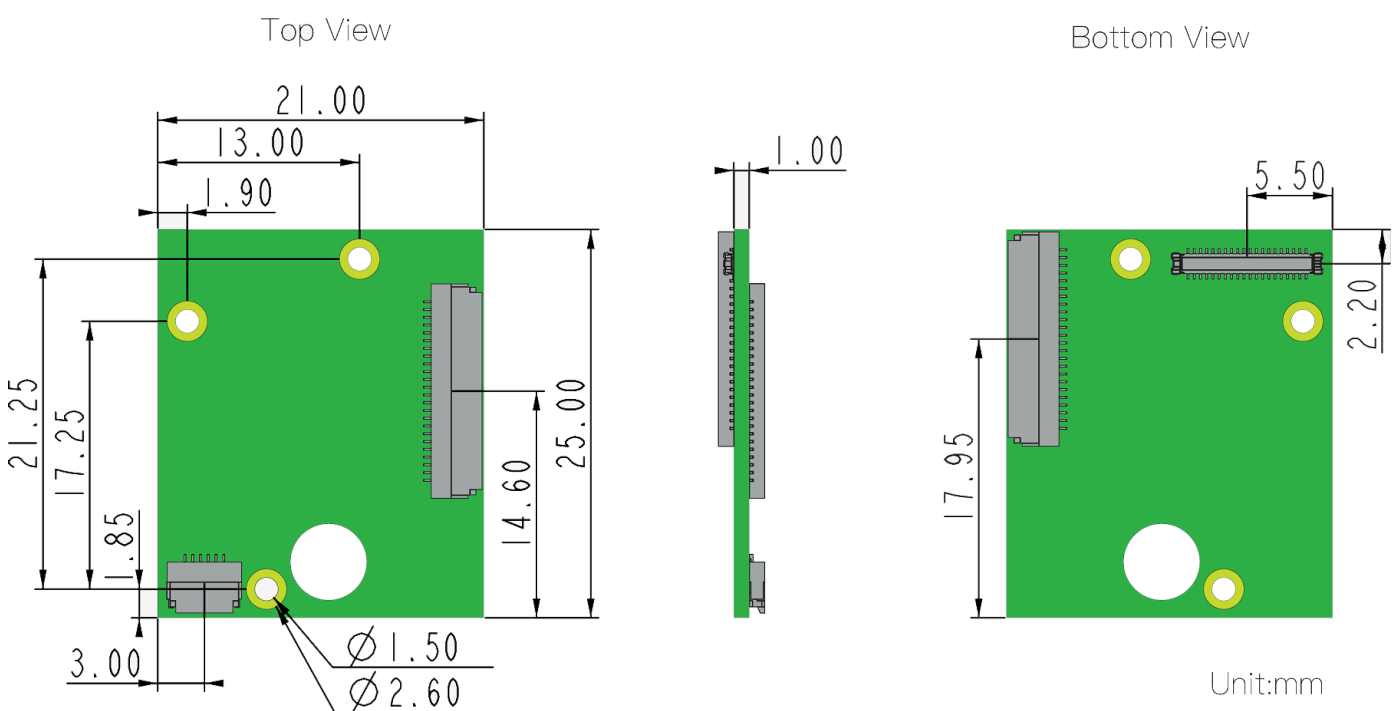


Figure 3: RAK14000 Module Mechanical Characteristics

Schematic Diagram

Figure 4 shows the schematic diagram of 40-pin WisConnector:

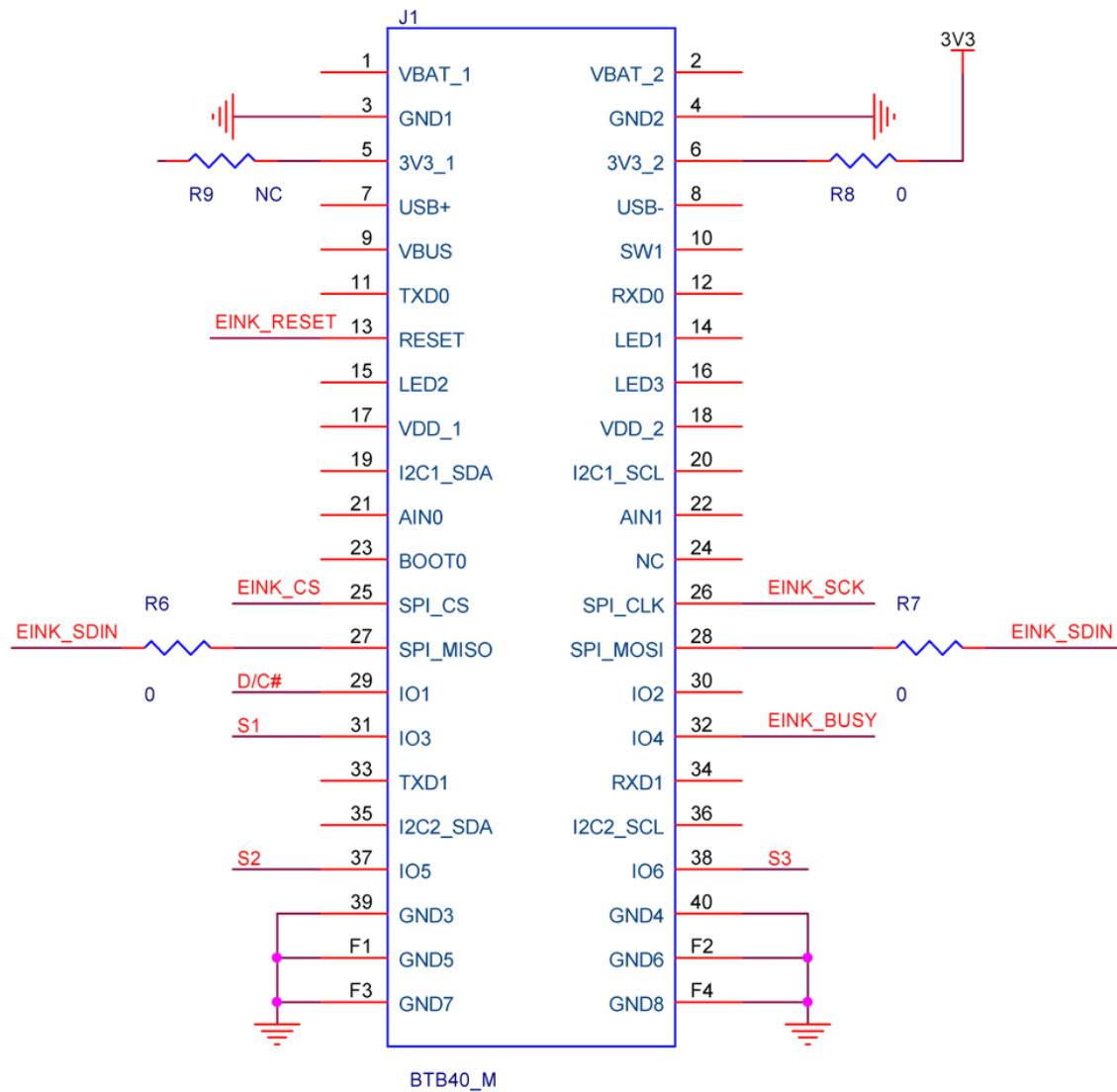


Figure 4: RAK14000 40-pin WisConnector Connection Schematic

Figure 5 shows the E-Ink Display Connector Schematic:

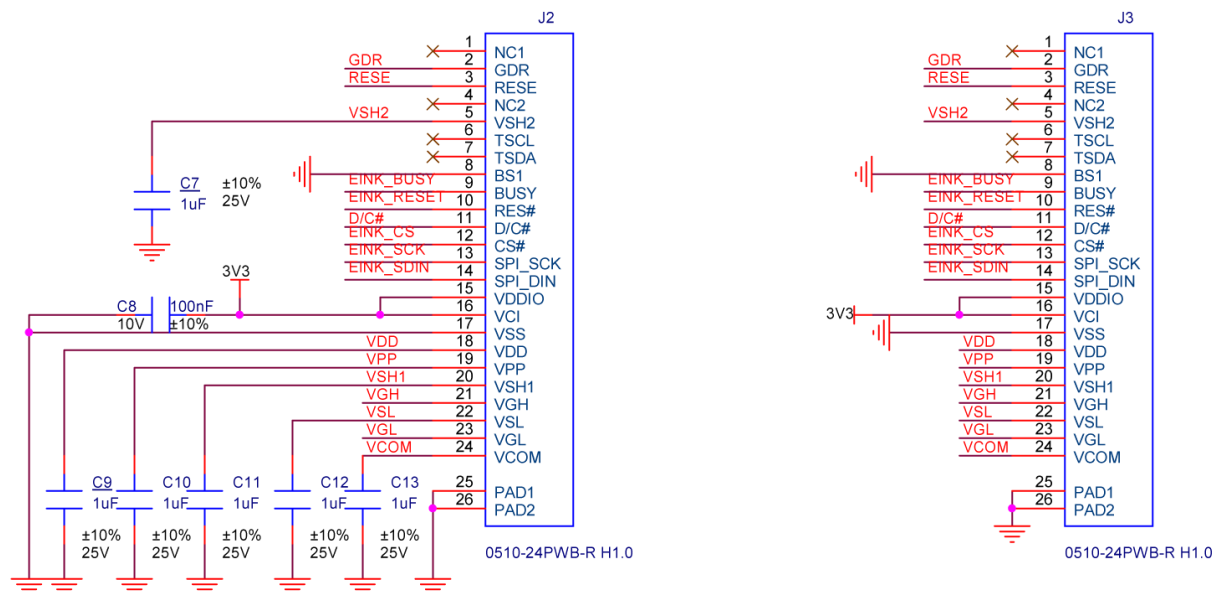


Figure 5: RAK14000 E-Ink Display Connection Schematic

Figure 6 shows the E-Ink Display Driver Schematic:

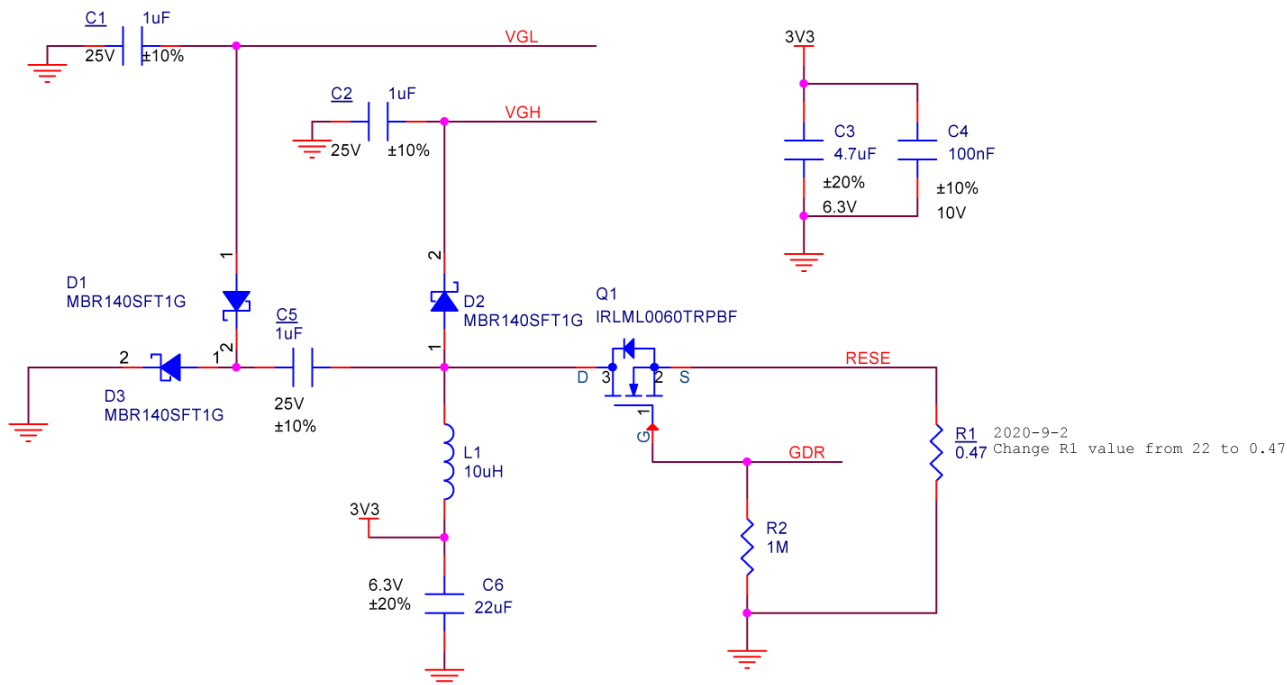


Figure 6: RAK14000 E-Ink Display Driver Schematic

Figure 7 shows the Three-Button Module Schematic, connected through FPC (Flexible PCB) cable:

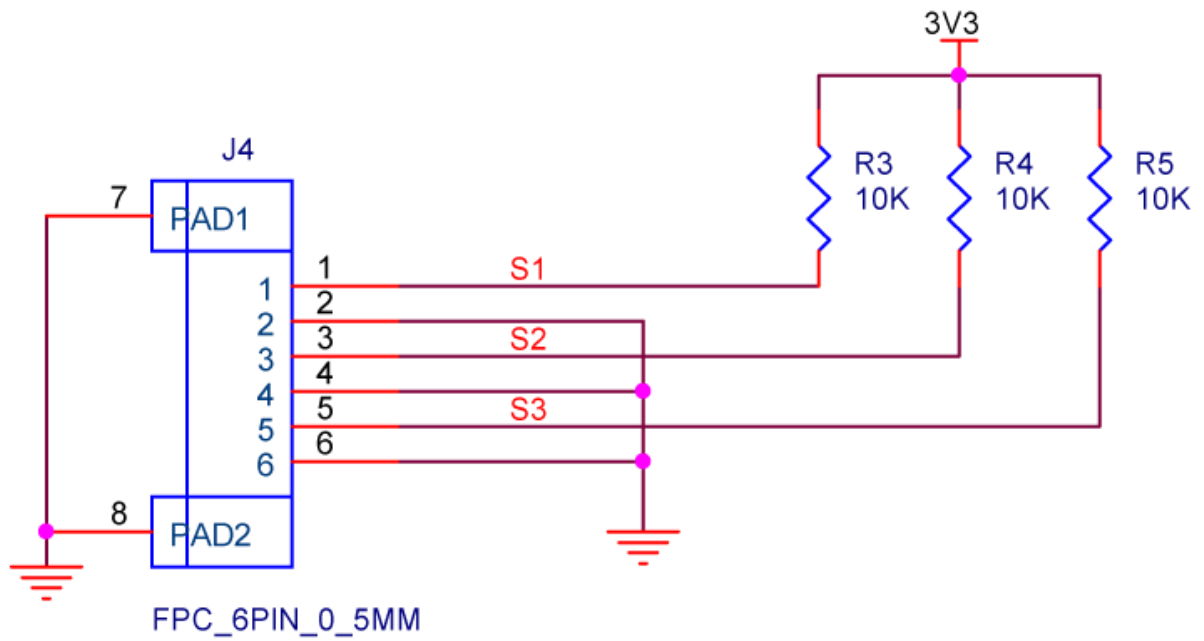


Figure 7: RAK14000 Three-Button Module Schematic